

THE EVOLUTION OF LANGUAGE AS CO - ADAPTATION OF HUMAN COGNITION AND PHYSIOLOGY(a perspective)

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Abstract

The article presents a vision of language and its evolution somewhat deviating from the focus in current language evolution research. First, although language is a systematic way of curating and encoding human semiosis, structure is focused on as the emblematic feature, while meaning is an afterthought. The focus on structure informs inquiries of bio-cognitive representation of language in the human organism, which predominantly features the brain and its cognitive functions. Consequently these preliminary assumptions frame inquiries of language evolution. In contrast the present article views language primarily as an expression of human concepts. In addition, here the language user is understood as body and mind and the contribution of human physiology in the materialization of cognitive processing in linguistic messages is viewed as an important contributing factor in shaping the language system and its evolutionary path.

The discussion starts by emphasizing that humans have evolved vast cognitive capacity which allows for detailed conceptualization of reality. And as social species we have evolved to share information by encoding it in signs, which engages human physiology for their material expression. That said, human physiology has not evolved in tandem with cognition, placing limitations on speed and efficiency of information transfer. The disparity between cognition and physiology, illustrated by the constraints of multimodal communication in encoding and articulation of the rich semiotic capacities of pre-linguistic minds is here understood as resolved by the evolution of language, a highly complex, compact and efficient system, externalized in a range of material representations as speech, manual sign, writing. The evolution of language is here viewed as a process of mutual adaptation of the extensive conceptual arsenal and its material expression in linguistic forms by incorporating three subprocesses : a. evolution of symbols from motivated signs, b. evolution of speech from multimodal communication, c. emergence of grammar for efficient packaging of meanings. In addition, the human organism has evolved physiological and cognitive adaptations, e.g. extended memory storage and speech capacities while human cultures have invented writing systems to accommodate the need for efficient and precise encoding of human semiosis.

By addressing viewpoints currently unexplored the article aims to contribute towards a more detailed and profound understanding of the evolution of language.

Keywords: evolution of language, symbolization, grammar, writing, semiosis,

Introduction

Language is a hierarchy of symbols, integrating meaning and structure, articulated in material form. Most prominent linguistic paradigms select structure as its defining feature and the evolution of language is understood as transformation from protolanguage, a crude representation of the essentials of human semiosis by a meaning-based communication system,

into a hierarchically organized structure-centred system of highly abstract grammatical concepts and their structural associations. Such crucial transformation is attributed by prominent biolinguistic paradigms (Bickerton 1984 and elsewhere) to a felicitous transformation in the human brain, resulting in the formation of a language capacity, a cognitive algorithm with highly specific properties, Universal Grammar, which, by its felicitous coordination with protolanguage, produces the semantic and structural complexity of language as we know it today. Such cognitive transformation is identified by the generative-biolinguistic context as the cornerstone of human speciation, resulting in the transformation of protolanguage speakers into speakers of modern languages, although understood as unmotivated by biological existence and even counter-adaptive, thus, in violation of evolutionary principles.

Alternatively, in usage-based context the systematic representation of meanings in linguistic forms of various levels of abstraction and structural configurations is defined as communicative technology, produced and periodically altered in reflection of communities' communicative preferences for effective communication (Givon, 1979 and elsewhere, Hopper, Traugott, 1993 to name just a few)

In the present article language is understood as a system of signs, i.e. a vehicle for meaning representation, and this function is featured as sine qua non in defining language and understanding language evolution. Crucially, human evolution is understood as imbalance between human cognition and physiology, i.e. the human brain has evolved enormous cognitive capacity to absorb, retain and creatively use information, disproportionate to the physiological capacities of the human body to externalize it in signs. Explanations for this disproportion is beyond the limitations of this article.

Significantly, representation and communication of the essentials of human semiosis, limited in scope and detail, are understood here as internalized by a form of a language faculty, given their adaptive significance in pre-civilization environments. The formation of structural complexity of language as we know it today is understood as communicative technology invented in attempt to resolve the discrepancy between the enormous cognitive capacities of the human brain to represent to a minute detail human semiosis, and the limitations of human physiology to cope with its externalization in communication. In sum, language evolution is understood as a solution to the disparity between human cognition and physiology as an obstacle to the temptation of humans as social animals to share thoughts and experiences with conspecifics.

1. The unique disparity of human evolution

1.1. The cognitive power of the human brain

Human speciation is understood in terms of extended intelligence, anatomically and physiologically reflected in increase in brain volume. And although encephalization in humans is a natural continuation of a process started in primate species, the human brain is a highly unusual evolutionary product not only in terms of volume and weight, close to 1,400 cc(cubic centimetres), but also as a percentage of body mass i.e. the adult human brain is 2 % of the

overall body weight.

That said, bigger brains don't necessarily suggest higher intelligence. In humans enlargement brings alteration in the brain connectivity, so a larger brain is a differently organized and as a consequence, a differently functioning brain (Deacon 1997 and elsewhere). Thus, human brain is not just quantitatively larger ape brain, it is a differently organized brain with the following unique anatomy :

- a. Large body/brain ratio, a result of general encephalization during primate evolution.
 - b. Significantly larger forebrain and the cerebellum in comparison to non-human primate brains, a result of selective increase.
 - c. Hemisphere asymmetries in the morphology of the two hemispheres.
 - d. Asymmetries of functions: lateralization of the brain functions, language-related capacities are concentrated mainly, although not exclusively, in the left hemisphere (Friederici 2018)
- Thus, “Human brains are not just large ape brains, they are ape brains with some rather significant alterations of proportions and relationships between the parts.” (Deacon, 1997, p. 255)

Most significantly, the human brain is an enormously powerful bio-cognitive entity, e.g. the average adult brain can store in memory some 50,000 linguistic items. This enormous cognitive potential is applied for extending knowledge and encasing it in signs.

“What is limited to humans in this regard is a large quantitative difference in the degree of complexity of our semantic universe” (Schoenemann, 1999, p. 312).

In sum, the hallmark of human species is the unparalleled cognitive power of the human brain, focussed on extending human semiosis.

From a different angle, although increased cognitive power is pointed at as crucial for the evolutionary success of our species, some doubt its adaptive advantages and argue that complex cognitive activity has high energy cost: the human brain consumes 20% of the total energy intake, which makes it highly inefficient. With these facts it is difficult to explain the need to evolve a highly inefficient way to survive, task, which other species accomplish for less.

“... Because the cost of maintaining a large brain is so great, it is intrinsically unlikely that large brains will evolve simply because they can. Large brains will evolve only when the selection factor in their favour is sufficient to overcome the steep cost gradient... evolution is the outcome of a balance between costs and benefits.” (Dunbar, 1998, p. 178)

And the evolutionary success of our species as the only survivors of the Homo branch suggests that the benefits outweigh the costs.

1.2. Evolution of human anatomy and physiology

The human body also has evolved distinctive anatomy and physiology: we are bipedal, hairless apes with unusual proportions of the length of arms and legs, highly agile thumb and a uniquely shaped skull housing an oversized brain (Mayr, 2001 chapter 11). To note, the distinct

anatomy the human hand has contributed to the development of material culture (Chavez, Morrell, 2022). That said, we share most of our genes, the basic body plan and functions, including brain functions, with the rest of the species and our anatomy and physiology offers a clear demonstration for continuity between humans and the rest of life forms.

But human physiology has not evolved in parallel with cognition, as the fact that we are skilled artisans is due more to our cognitive power than to enhanced physiological flexibility of the hands.

At the same time bipedal locomotion and the bodily adaptation to it is viewed by scholars as the bio-physiological evolutionary development driving human cognitive evolution. Bipedalism is attributed a prominent role in human physiological and cognitive uniqueness, by marking a dividing line between animals and primates and as an initial step towards humanity, starting from Australopithecus. Most scholars converge on the view that bipedalism and the restructuring of the human skeleton as a result of it, is the main cause for the shortening of the intrauterine development schedule for the fetus growth. Ragir has argued that the fact that human babies are born physically and cognitively premature and in need of care for a long period after birth, compared to babies of mammal species, e.g. elephants, wolves, etc. which become functional soon after birth, is the cause of all the major developments in human evolution with crucial biological and cognitive consequences for encephalization and development, e.g. prolonged childhood, emergence of adolescence as a new and unique life stage, in addition to social consequences, e.g. formation of the family unit responsible for the care for the immature child, and other social structures which resulted in evolution of capacities for linguistic communication (Ragir, 2001 and elsewhere). Similar arguments are made in earlier publications by Aiello (1996).

That said, a number of non-human Homo species, known for various degrees of bipedal waking, starting from Australopithecus, are not known to have had any form of language and linguistic abilities of Erectus are for now speculative. So, the role of bipedalism in the evolution of human linguistic abilities is only remote and indirect.

1.3. Human evolution as disparity between cognition and physiology

The defining feature of the human species is the brain, the most powerful cognitive entity nature has invented. Human evolution is mainly an evolution of the brain and cognition. Brain size and functions are the fundamental aspect of human speciation. At the same time, human bodies are not very different from ape's and even lower species genetically, metabolically, physiologically.

One may argue that humans have evolved fast running abilities, superior to many mammal species and highly adaptive in hunting. That said, although fast running has been an advantage in human hunters' ability to kill large game, it is most definitely our cognitive skills, e.g. ability to plan, organize and make quick decisions, the major contributor to human evolutionary success. And although humans are known for dexterity as we are highly skilled artisans, as a matter of observation, many species, from squirrels to monkeys, rival us in the range and precision of movement in their four limbs. The most significant physiological difference in

humans is in speech capacities. In this sense Lieberman(2016) has argued that basal ganglia have a fundamental role in both cognitive and motor tasks and brain damage in this area disrupts both activities, e.g. in patients with Parkinson disease. Moreover, “ basal ganglia play a critical role in associative learning and in planning and executing motor acts including speech in humans and songs in songbirds” (Lieberman *ibid.* p.138) pointing at evolutionary continuity of this part of the brain architecture. This, though, does not negate the fact of unbalance between human cognition and physiology.

The realization of the human physiological limitations, disproportionate to our cognitive abilities, have prompted the domestication of some animal species, e.g. horses, dogs, onto which physiological demands are offloaded e.g. in hunting, transportation, etc. Human civilization deals with this unbalance by introducing technological innovations. In short, continuity with the rest of life forms is much more pronounced in human biology, while human cognition displays clear dissimilarities. Thus, human cognition and physiology have had separate and divergent evolutionary paths.

The evolutionary innovations in human development reflect the demands of cognition by affording a period of adolescence, predominantly consisting of channeling and shaping this cognitive power by extended learning and practice of skills.

Human thoughts are communicated by human physiology. In spoken dialogues, the most clear illustration of natural language in use, linguistic utterances are accompanied by non-linguistic multimodal body signals, e.g. manual gesticulations, body posture, vocalizations, facial expressions etc. which clarify and complement the intended message.

That said, the stream of thought is much faster than the human physiology can convey, which overwhelms the speech capacities, even if complemented by non-linguistic signals, given the time constraints of a conversation. And one can comprehend many more concepts than one can verbalize, i.e. linguistic comprehension is more extensive than linguistic production. Moreover, the communicative use of human physiology, especially in sign languages, imposes high energy demands on the body, suggesting an explanation for why speech, a much less energy-demanding physiological activity, has evolved as the universal default modality, while sign languages are used only when speech organs are damaged. And assuming pre-human multimodal communication as a predecessor of language (Slocombe, Waller and Liebal 2011) the high energy demands it imposes on physiology makes it inefficient vehicle for making the fast stream of human thought explicit. In this context one is prompted to speculate that the disparity between cognition and communication has created a bottleneck, the solution of which is at the root of the evolution of language. It is envisioned here as a three-fold process, encompassing both alterations in the individual cognitive and physiological potential and in the social and cultural organization of communities. One, human anatomy and physiology evolved speech capacities as high -speed energy-efficient vehicle for communicating human thoughts by phylogenesis. Another is restructuring in the system of signs by evolving symbols from motivated signs as a way of encoding more concepts with more precision and accuracy. The third is the formation of complex grammatical systems as a condensed packaging of concepts most frequently used.

2. Language, a semiotic perspective : language as a meaning-making system

2.1. Language as a system of signs

Signs are formed for the purpose of representing and conveying meaning, i.e. the very presence of signs implies meaning. Language is a system of symbols, a type of signs. Ever since Saussure linguistic forms are defined by their symbolic nature. Within the usage-based perspectives language is defined as lexicon of constructions, i.e. accidental associations of form and meaning of various types and sizes, thus, by definition, a symbolic system. And although in generative context, where language is defined as computation and the focus is on structure, computation is by definition manipulation of symbols, which implies meaning representation, justifiable only with the purpose of communication. Thus, whether explicitly recognized or not, language is a symbolic system where meanings are encased in linguistic forms.

Linguistic meanings are materialized by a variety of material avenues: speech, gesture, writing. Moreover, the very fact that language is a symbolic system implies that it is inherently a system for communication as the only reason for the existence of symbols (and of any signs) as labels for concepts, is to make these available to an external entity, i.e. with the only purpose of communication. Thus, language as a system of symbols is inherently designed for communication.

1.2. Language as integrated system of meaning and structure

In generative context formal theories of language define it in terms of integration of two partially independent and differently organized systems, lexicon and grammar, i.e. meaning and structure. Otherwise, in usage-based approaches relying on empirical findings, language is an integrated system where lexical and grammatical forms form a continuum. The degree of membership in a category is defined based on similarity to the prototype, or the best example for a category of entities. For example, the distinction of regular and irregular verbs is a matter of degree as some verb forms follow regular patterns while others deviate from them to various degrees, e.g. hit, hits, but hit vs. hit (past). In addition the lexical categories verb, adjective and noun are points in a continuum based on their semantic representation of time-stability. At one extreme of the continuum are active verbs which denote rapid change in a state, e.g. jump, start, begin, etc. States of short duration are usually lexicalized as verbs, e.g. initiate, activate, etc. adjectives e.g. young, intense, and some nouns, e.g. youth, activity, while permanent states are lexicalized as nouns, e.g. man, woman, sky, tree, establishment, etc. and they form the opposite extreme of the continuum. Continuity in the semantics of grammatical categories is along the continuum from content nouns to forms with increasingly more abstract meanings, i.e. prepositions, tense/aspect/mode markers, definite/indefinite articles, etc. (Givón 1979, p. 14-). Thus, language is organized along a continuum of meanings. Moreover, forms of some level of abstraction, e.g. prepositions, depend for their existence on content words, e.g. tense/modality/aspect morphology etc. are conditioned upon the existence of lexical verbs

suggesting the internal integration of the system as a whole. The gradient nature of language is also pointed out by Greenberg(1963) who understands universals as organized in clines. Lexical items can change their position on the cline by undergoing a process of grammaticalization (Hopper, Traugott, 1993). Comrie and Kuteva (2005) have demonstrated that concepts usually encoded in grammatical forms in modern languages almost always can alternatively be expressed in lexical words , pointing at synonymy between lexical and grammatical forms. Some prominent generativists have reached similar conclusions (Cullicover, Jackendoff, 2005; Jackendoff, Wittenberg, 2014).

Significantly, the lexicon is defined not as a sum of independently existing words, selected out of a pool to fill specific empty places on predetermined syntactic trees, as defined by the generative paradigm, but as a system in its own right with its “ semantic topology”. In addition, an item in the lexicon is designed to be a part of a structure, that is, a component of a hierarchy. All types of linguistic entities, from lexical word to a definite article, are not meant to be used in isolation but as a member of a lexicon, an internally organized system in its own right, where each member exists in semantic relations to other members and occupies a certain unique semantic space, e.g. semantic spaces are interconnected and from relations of synonymy, antonymy, hyponymy, and semantic frames (Fillmore, 1976).

Importantly, the systemic nature of language, based on semantic relations, is foundational for syntax. A linguistic form has grammatical features which, in structuralist terms are termed as paradigmatic features (defining the membership in a grammatical category) , and syntagmatic features (defining the potential membership in larger hierarchically organized abstract structures,i.e. phrases and sentences. Thus, a linguistic form contains information which points at its potential combinatorial options. To note, the potential for combinatoriality is implied as part of the definition of any linguistic entity as even in the simplest forms of language , pidgins, early child language etc. words in their bare forms are used in meaning-based combinations. In sum, language is an integrated system where meaning and structure presuppose one another.

2.3. Language processing and its integration in the brain

In generative/biolinguistic context the brain's specialization for language processing is centred on grammar. As per the generative formalism and its implementation from biolinguistic perspective the language processing is performed by the grammar algorithm (UG) on abstract grammatical categories. Meaning is here understood in terms of computation as abstract meanings organized into a Language of Thought (LOT) (Fodor 1975 and elsewhere) and instantiated in the human brain by a cognitive module in close association with UG, i.e. both meaning and structure in language are understood in terms of abstractions.

Nevertheless, Pulvermuller (1999 and elsewhere) demonstrates that the brain locates and processes linguistic entities according to their use in real communicative events. For example lexical words are represented in neuronal networks or semantic hubs by their meanings and meaning associations and categorized as denoting living entities (animals), man-made small objects, tools, etc, i.e. examples of actual experiences, not as abstractions. Moreover, concrete nouns are represented together with the perceptual stimuli (Pulvermuller, 2005) which

accompany their use, e.g. verbs are associated with motor neuronal programs, content words denoting odours, sounds, visual perception, etc. are represented in association with the respective perceptual neuronal programs. Thus lexical words are represented in the brain in association with the extralinguistic information reflecting their potential use in reference. Linguistic entities are categorized also by phonological criteria of length, e.g. long words, short words. Significantly, even abstract linguistic forms are represented as grounded in perceptual experiences (Pulvermuhler, 2018).

In sum, the brain categorizes and stores in its topography linguistic forms in terms of their practical use in real experiences, not by membership in abstract categories, and applies its own logic in language processing and learning, which is incompatible with the logic by which language is formalized in linguistic theories. Importantly, the representation of language in the brain and its processing and learning must be understood in the context of the overall architecture and functioning of the brain, its anatomy, bio-chemistry and connectivity, as it functions as an integrated unity where language processing is one among multiple other functions, e.g. perception, emotion, etc.

In this sense theories of language, which postulate modularized cognitive entities, e.g. UG inconsistent with the brain's internal logic of integration and functioning must be viewed as problematic and require revision as they create an unbridgeable gap between linguistic theory and brain sciences, a major stumbling stone for modern linguistics. We cannot understand language in isolation from the properties of the brain. On the other hand, understanding language as part of the architecture and functioning of the brain would contribute to a more detailed understanding of the brain itself.

2.4. Linguistic symbols are grounded in experience

Although language is traditionally defined as structure the existence of structure cannot be justified without its contribution to meaning, i.e. meaning representation is the *sine qua non* of language. To remind, in generative/biolinguistic context meanings of linguistic symbols are understood in terms of abstract computations, pertaining to Language of Thought, i.e. independent of human experiences (Fodor 1975 and elsewhere).

That said, linguistic meanings cannot be divorced from the reality where humans exist and function, i.e. linguistic forms must reflect extralinguistic reality to be meaningful which determines its usability. In this context one must reinterpret language as a systematic way for representing human semiosis for the purposes of communication. Moreover, we learn, understand and articulate a significant part of human semiosis through encoding concepts in linguistic forms. In this sense the structure of language cannot be independent of the structure of human thought which, itself is a reflection of the outside world through human perception and conceptualization. This interdependence of concrete experiences, reasoning, and language is reflected in the brain has evolved to represent linguistic items by their meanings and their potential use in communication in a unique capacity for symbolic reference, (Deacon 1997 and elsewhere), a complex combination of various types of referential relationships among symbols as members of a symbolic system, between a symbol and its referent, and among

referents as objects in reality. In this sense, the systemic relationships of linguistic forms correspond to the relatedness of things in the world. Importantly, the dual interdependence between the system of linguistic symbols and the perception of the natural organization of the world is fundamental as the defining feature of language in its role in communication.

Conceptualization of reality is species-specific in reflection of species' unique biology and cognition which determines their interaction with the environment. Human concepts can only be formed by human bodies and minds. Some concepts are universally encoded in all languages as they reflect the very basics of human experience, as pertinent to essential survival and form the fundamentals of human semiosis. Wierzbicka (1992 and elsewhere) and Goddard (2007) outline the following:

- a.** material reality from anthropocentric perspective : topography, flora and fauna, climate, including human perspective of size, distance, speed, time,
- b.** human body and mind: body parts, internal organs, cognitive processes, physical activities, nutrition,
- c.** human interactions and relations, : kin, social stratification,
- d.** artifacts of human life: housing, clothing.

In addition, human communities conceptualize their invented reality , a reflection of ethnic, cultural and social idiosyncrasies.

- a.** culture: religion, cultural beliefs and activities, cultural values and life style,
- b.** civilization : agriculture, science, philosophy,
- c.** social organization : types of social organization and social institutions

The linguistic representations of these meanings include all types of linguistic forms of all lexical and grammatical categories. Importantly, languages lacking most of the highly abstract categorical and structural sophistication of Universal Grammar, e.g. Piraha, Riau, etc. have proven to be equally capable of representing human semiosis.

The understanding of language as meaning-based is articulated by Cognitive Grammar, one of a multitude of theoretical perspectives embracing a usage-based approach (Langacker, 2008, among others). And in the context of Cognitive grammar the distinction between grammar and lexicon originates in conceptual structure with the distinction between Conceptual Structuring System, which “ provides the structure, skeleton...of a given scene” (Evans, Green, 2006, p. 192,) and Conceptual Content System, a token which “ provides the rich detail of a particular scene..”(ibid. p.192) Although this structural distinction at conceptual level is relatively vague, it provides the foundation for the formation of the semantic system by introducing thematic roles e.g. agent, theme, experiencer, place, manner, etc. identifying who does what to whom, how, where, when, etc. and their associations as structural foundation of grammar. So, grammar arises out of internal reorganization of conceptual structure and the formation of semantic roles as intermediate level, or interface, facilitating the encoding of propositions in sentences. Language is, therefore, better understood as a meaning-making system given that the organization of the material world is reflected in the organization of the conceptual system, which, in turn, is reflected in the semantics and ultimately grammar.

In Cognitive Linguistics grammatical categories, e.g. noun, verb, participle, preposition, etc. are defined in terms of different ways of conceptualizing human experiences, e.g. nouns as prototypical representation of material objects of stable state, verbs as encapsulating processes

as fluid interactions of participating entities, prepositions as spatial, temporal relations between objects as stable entities, etc. Modal verbs encapsulate the psychological aspects of human experiences by encoding speakers' attitudes. Moreover, structural relations are determined by semantic features, e.g. differences in meaning determine the differences in structural configurations between mass nouns and count nouns, and between verbs, participles, gerunds, infinitives, etc. In short, given that concepts are organized into structured combinations, these are reflected in linguistic categories and their roles in the organization of language, i.e. grammar is meaningful (Langacker, 2008). Similarly, Schoenemann (2005) argues that the structured nature of language reflects the structured nature of extralinguistic reality which is reflected in its structured cognitive representation by the human mind.

“Given that the world is hierarchically organized (or at the very least that we have evolved to conceptualize the world hierarchically) should we not expect the syntax of our languages to reflect this?” (Schoenemann 1999, p.330)

That is, the hierarchical structure of sentences is a reflection of the way conceptual structure is organized, phrase structures are groupings based on meaning relations, and syntactic structures represented in the human mind are derivative from the universal cognitive capacity of humans to conceptualize reality in complex ways.

3. Meaning representation and language use

Humans like routines, human behaviour is patterned. The life of the human individual is structured along his/her roles in 3 major types of social interactions, i.e. as a professional in interactions with colleagues, as a member of a family, neighbourhood, social circle, and as occasional interactions with strangers, i.e., fellow humans. These 3 types of communicative interactions determine the ways one organizes one's linguistic behaviour, i.e. one's E-language. And although in modern linguistic thought the linguistic behaviour/E-language/parole of the average human is assumed to be an unstructured mix of various types of utterances, one cannot but notice that it reflects the patterns of human interactions.

Every human speaker uses language in 3 different ways : **1.** at a professional level as a professional tool for precise and detailed explication of complex ideas in science and technology, for defending and/or debunking arguments in philosophical debates and court rooms, etc., **2.** at a personal level as a vehicle for social contacts and **3.** as a tool of last resort for occasional interactions with fellow humans in unusual communicative circumstances, e.g. with foreign tourists, etc.

And if one assumes, in the context of the usage-based paradigm, that the language system is shaped by linguistic behaviour, it follows that the 3 types of linguistic behaviour determine the corresponding types of language subsystems. The linguistic behaviour of the average adult speaker, or E-language, is internally diverse system organized into three subsystems which co-exist in parallel. Each of the three types, which I have labeled as “code systems,” “inferential systems” and “rudimentary” systems, are systematic in their own right as they display distinct internal organization around different sets of principles. To note, the structured nature of E-language is not a hypothesis, it is a fact, easily verifiable by self-monitoring,

although so far not noticed by linguistic paradigms. Each of these represent human semiosis in different ways. The language system has evolved diversified opportunities to communicate meaning in response to the diverse demands for information in the aforementioned 3 types of human interactions.

2.1.The code subsystem

A code in general can be identified as composed of signs, usually symbols, each a unique representation of a concept and its potential participation in combination with other symbols. Examples of code systems are the Morse code, artificial languages, traffic signs, etc. Code systems are usually designed for communicating specific types of information, i.e. both the signs and the rules reflect a very specific communicative context, e.g. the traffic signs are designed for traffic, the genetic code is designed by evolution to encode the basic instructions for the formation of a biological body, artificial languages are designed for specific programs, etc. Language, as defined by formal perspectives, bares the characteristics of a code, and as such displays the following characteristics:

- a.** It is composed of linguistic primitives, members of a lexicon, as one-to-one stable associations of a meaning and a form, i.e. synonymy and homonymy is non-existent.
- b.** These are defined by their membership in discrete and well defined grammatical categories.
- c.** The so defined linguistic primitives are organized into sentences following principles of grammaticality.
- d.** The meaning of a sentence is the sum total of the meanings of the composing words and their place in the architecture of the sentence.
- e.** The formal structure of a sentence is highly detailed, with highly abstract grammatical forms and unlimited debt of recursion of phrases and sentences.
- f.** A sentence encodes a complete proposition . Explicit and complete mapping between semantic structure and grammar is the norm. All thematic roles in the theta grid of a verb are expressed in grammatical categories. The agent consistently occupies the subject position in the sentence structure.
- g.** The code-like system is self-contained, independent of context of use. This facilitates the correct decoding of the meaning by people with vastly different experiences and views.
- h.** The message for the sender and the receiver are identical.
- i.** The function of code systems is mainly to inform, so sentences are mostly statements.

Language as a code is used by highly educated language-trained professionals for whom language is a working tool and is most often materialized in writing. Its role is to articulate and disseminate in a clear and concise manner complex and timeless ideas among communicators separated by space and time. It is, in addition, organized by the spelling and punctuation conventions of the respective writing systems. Thus, the code system is designed in response to

the demands of complex civilization. And although the code system encodes a huge amount of information, it has its limitations as it is inefficient in encoding human emotions, attitudes, context details and social and cultural idiosyncrasies.

2.2. The inferential subsystem

Language as inferential system is designed for use in informal unplanned communicative interactions and reflects the strengths and limitations of communication by direct linguistic interactions in dialogues. And although generally ignored by standard theories as unsystematic and, therefore unworthy of serious scientific study, the inferential language systems are systematic in their own way and display their own unique characteristics which are universal as follows:

- a.** The inferential system has information-based, not structure-based internal organization as it is organized around information structure (topic vs. focus).
 - b.** It is designed to reflect the biological demands of the modality of speech and manual signs, e.g. when used in speech intonation assumes some grammatical functions, as in the expression of questions without the use of question words.
 - c.** The building blocks of the system are flexible associations of form and meaning as standard meanings are interpreted with context-dependent flexibility.
 - d.** Grammatically defective elements of unclear syntactic features, unclear morphological class and irregular phonology, or in Jackendoff's terms 'defective items' (Jackendoff R.2002), 'mm', 'wow', 'sht' also abound. Expletives are frequently used.
 - e.** Elliptical and abbreviated forms abound. Formulaic phrases are often used. Semantically vague words and phrases, e.g. 'that fellow', 'that thing', 'people' are often used.
 - f.** Linguistic expressions form utterances.
 - g.** Most utterances contain the most frugal use of constructions, absolutely necessary for making the message understandable, which in the context of Universal Grammar would be described as fragments of phrases and sentences. Most verbs have incomplete argument structure with only a single argument.
 - h.** When an utterance is a full sentence the order of the elements is flexible to signal speaker's attitude.
 - i.** The inferential language system uses grammatical forms with clear contribution to meaning, e.g. markers of plurality, modality, tense, aspect markers, case markers in languages with detailed case systems (German, Russian, etc) as required by the context. That said, omission of grammatical markers which do not contribute to meaning and have only structural values, e.g. definite and indefinite articles in English, is one of the most notable characteristics.
 - j.** Although in the context of the generative formalism such less than full applications of Universal Grammar are considered structural deficiencies, these do not result in communication disturbances, as despite these apparent structural gaps the complete meaning of the utterance is successfully recovered from the context.
 - k.** Small clauses, almost complete lack of embedding of phrases and sentences is the norm. Sentence coordination is preferred, subordination is rare.
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- l.** The meaning of a sentence is different from the meaning of the utterance and the difference between the two cannot be stipulated in advance by a code-like rules. The meaning of an utterance is calculated as the meaning of a sentence and the speaker's communicative intentions, or illocutionary force (Austin, J.1975). Utterances communicate the intended meaning in addition to the speaker's attitudes.
- m.** The meaning intended by the sender is most often different from the meaning understood by the receiver. An inferential system is based on the assumption that participants are individualities with different minds and different life experiences in different communicative circumstances , which creates the potential for different interpretations of the same linguistic forms.
- n.** Utterances form part of spontaneous dialogues constructed by universal principles of cooperation in communication and general maxims of interpretation, outlined by Paul Grice . The meaning of individual utterances within a dialogue are interpreted by principles of relevance (Wilson, 1998 and elsewhere). Such system is used during relaxed, casual conversations among people with close social ties with individuals united by personal relations and common life style, e.g. family members, neighbours, villagers, which presupposes shared knowledge, cultural values and life style. It is situation-dependent, routinely accompanied by non-linguistic communication as major contribution to the understanding of the message. The inferential system is universally used by all human communicators, regardless of education, age, gender, ethnicity, social status or profession. It is a system is designed to be used in conjunction with non-linguistic signals the human organism sends , intentionally or by impulsive reaction, e.g. facial expression, pointing, body movements, etc. It has its limitations as , given the direct interaction of communicators, implying common knowledge of cultural values, social norms, material circumstances, a good part of information is either shared, eliminating the demand for communication, or /and communicated by non-linguistic means. Thus, the inferential system has its own limitations as the use of linguistic forms covers only a limited portion of the information exchanged.

2.3.The rudimentary subsystems

The term “ rudimentary “ is descriptive of a number of language systems with similar structural properties and communicative functions, usually described as “ protolanguage “ (Bickerton 1990) and characterized by predominant use of lexical words and minimal use of basic grammatical categories for marking objects, actions, properties etc.. From the standpoint of modern languages one can talk of highly simplified grammar. And although this type of linguistic communication is characterized by Bickerton as grammarless, as it indeed displays minimal use of abstract grammatical forms, it is by no means structureless. The internal organization bares the hallmarks of a human language, i.e. based on conceptual structures encoded in linguistic forms, mediated by theta roles and semantic structures, and fully capable to encode and communicate infinite human thoughts by finite linguistic means. Given that the lexicon under the definition by Construction grammar (Goldberg, 2003) includes all types of linguistic forms, from content words to morphemes, to phrases, the preferential use of one type

over another does not justify the label protolanguage or pre-language, but, in fact, eliminates the need for the theoretical distinction of protolanguage and language proper.

The following characteristics are emblematic of rudimentary systems :

- a.** a small vocabulary of lexical words with concrete meanings, organized in grammatical categories of object words (nouns) and action words (verbs),
- b.** extensive use of compounding,
- c.** serial verb constructions, sentence embedding is largely absent
- d.** hierarchical structure based on semantic relations,
- e.** morphologically simple lexical forms, i.e. bare stems are juxtaposed,
- f.** a very limited number of grammatical words with more or less abstract meanings, little or no morphology,
- g.** absence of abstract grammatical categories of subject, direct and indirect object, case, tense, aspect, complementizer, characteristic of grammars of modern languages,
- h.** no grammatical forms to express negation and questions, replaced by intonation,
- i.** no signs of grammaticalization process,
- g.** no fixed phrase structure and phrase embedding
- k.** one-place predicates,
- l.** extensive use of stress and intonation as a replacement of grammatical devices

A number of communication systems share these characteristics:

- A.** the linguistic output of small children during the initial stages of language learning /acquisition, who learn language under normal circumstances,
- B.** the linguistic output of feral children who learn language under abnormal circumstances,
- C.** newly emerged sign language of Al-Sayyid, which is language formation in early stages of the process ,
- D.** basic variety (Klein, Perdue,1997)
- E.** trade pidgins,
- F.** the linguistic production of agrammatic aphasics,
- G.** the linguistic production of trained apes.

Naturally, the amount and scope of information communicated by the rudimentary systems is highly restricted, which justifies the liberal use of non-linguistic forms of communication : gesticulations, facial expressions, non-linguistic vocalizations, etc. which accompany and complement the linguistic message.

The structural properties of the rudimentary systems are tailored to their function as they all cover the most essential and primitive information needs of modern humans in highly unusual

communicative circumstances , e.g. in spurious short interactions among speakers of different, mutually unintelligible languages, as in conversations with foreign tourists and in primitive trade exchanges , in communication of cognitively impaired and/or developmentally challenged individuals, in initial stages of language development by children, in initial stages of the formation of new languages, in some isolated examples of linguistic communication by non-human species. In short, rudimentary systems are designed to cover only the very basics of human semiosis.

In sum, the language system, understood as a combination of 3 subsystems, is tailored to fit human demands for information in the 3 types of communicative routines human are engaged in their daily lives. And although so far no linguistic paradigm recognizes the structured nature of E-language, its existence is a fact of life, easily verifiable by simple self-monitoring.

3. The human body, a vehicle for externalization of the language system

The seminal paper by Poizner, Bellugi and Klima (1990), which demonstrates that sign languages have properties highly similar to those of spoken languages, has been taken as a confirmation that the physiological properties of the channel by which the language system is externalized are of no consequence in shaping the structure of language.

That said, linguistic forms, from lexical words to grammatical morphemes, are largely symbols, i.e. a type of signs, which, by definition are stable associations of an idea and its material form for the purpose of communication. It is important to underscore, for the purposes of the present article, that linguistic signs can only function as designed if the language system is externalized through its material expression. One cannot exist without the other. And these two aspects of language, the abstract and the material, interact and influence one another. The language systems as a system based on signs, is made possible by the human individual: the abstract aspect by human cognition and the material aspect by human physiology, i.e by the human organism as a whole. And given that the production and interpretation of linguistic signs places energy demands, both cognitive and physiological, on the body, this suggests that human cognition, physiology and anatomy contribute in shaping the language system by influencing language use, thereby suggesting a direct co-dependence between the abstract language system and the material properties of channel of externalization. The language systems today are materialized by three material forms, speech, manual signs and writing and each interacts with the language system and influence it in different ways.

3.1. Bio-cognitive demands shape language subsystems

Casual conversations among individuals with close social ties occupy about 20% of all waking time of humans . Spoken dialogues is universally the most frequent form of language use (Dessalles, 2007). The human organism has the ability to process spoken language with remarkable speed and accuracy. A linguistic item is processed on average as follows: 65 milliseconds (msec) for the processing of a phonological form, 250 msec for processing the semantics, 1-2 sec for processing the grammatical properties of a sentence (Givon ,2002, p.

74). The demand for efficiency in articulation explains the fact that shorter forms are used with high frequency in all languages (Zipf 1949). Thus, the physiology of speech influences language use.

The processing of spoken language includes both production and comprehension and the biological organism imposes limitations on each as each participant is both a speaker and a listener and language processing places energy demands on both physiology and cognition. Because language use is a negotiation between language production and comprehension, there must be a middle ground where the demands of both communicators are reasonably satisfied. The spoken dialogue imposes relatively low cognitive and articulatory demands as it tolerates grammatical inaccuracies, conceptual vagueness, incomplete phrases, incomplete and often inaccurate pronunciation. Cognitive and articulatory deficiencies are tolerated as they are of little consequence for the accuracy of message comprehension. In addition, in dialogues the participants, often more than two, alternate in turn taking which minimizes the cognitive and articulatory effort even further as it is distributed.

Although casual dialogues are physiological activity of relatively low energy cost, narration in spoken form as long spoken monologues, e.g. speeches, lectures, etc. are rare communicative events because they cause physiological exhaustion for the speech organs. This places limits on the amount of information transferred given the time and space limitations imposed by the ephemeral nature of face-to-face communication.

The inherent limitations of speech and manual sign are complemented by the severe limitations of linguistic resources for the users of rudimentary language systems, who resort to elevated implementation of non-linguistic multimodal signals, e.g. facial expression, body posture, eye contact etc. to make themselves understandable, making the contribution of language only secondary.

While the physiological demands of speech production are minimal, one can reasonably anticipate that the production of sign gestures place much higher energy demands on physiology. The elevated kinetic demands of manual signs essentially prohibit the use of sign language in long speeches as this would lead to physical exhaustion. This naturally limits the use of sign language to short conversations. It also explains the fact that spoken and not sign language is the default channel and is only used when the vocal-auditory channel is unusable, as a solution of last resort, not as alternative to speech.

The use of language as a professional tool increases the cognitive demands on the brain beginning with the memorization and retention of a large vocabulary and careful selection of the most appropriate lexical and grammatical tools which requires focus, time-consuming deliberations and meticulous selection among alternative variants by the sender, driven by the demand for the detailed and precise exposition of sophisticated ideas by relying exclusively on language. Similarly high cognitive demands are imposed on the receiver by the need to evaluate the ideas of the author and compare and contrast to one's own philosophical, theoretical, political, etc. convictions. Thus, substantial cognitive demands are imposed on the brain by the demands for clear and detailed exposition and evaluation of complex ideas in linguistic terms by both author and audience.

The elevated complexity of the language system requires not only cognitive energy by both

sender and the interpreter, but also imposes elevated physiological demands for clear articulation on the speech organs which contributes to the overall energy demands on the body, e.g. in oral arguments in court, lectures, etc. The fact that the sound is ephemeral and fades away rapidly puts additional pressure on the memory of all participants. This explains why this type of activity is restricted to rare instances of language use, e.g., speeches are rare events, lectures and presentations are restricted in duration and television and radio shows, if monologues, are interspersed by breaks. The limitations of human cognition and physiology are also reflected in the structure of preliterate languages, where information is shared by oral traditions, e.g. by memorization and repetition of myths and tales, which restricts the speed and accuracy of information and imposes inferential structure on the language system.

The limitations of human body and mind in dissemination of knowledge prompt the invention and proliferation of writing where the stability and permanence of letters replaces speech as ephemeral medium and transfers the physiological effort to the visual system as a much more efficient use of physiological resources. In addition this significantly alleviates the cognitive pressure on the brain imposed by the immediacy of coding and interpreting of speech as written texts allow time for precision in encoding the ideas by the sender and accuracy in decoding by the receiver. In this way the cognitive effort of both participants is distributed over time allowing the dissemination of complex ideas without the investment of a great deal of cognitive energy. This allows for amplification of grammatical detail and sophistication in reliance on code-like language structure.

3.2. Human physiology shapes language use in speech, sign, writing

The overwhelming majority of linguistic interactions are conducted through the vocal-auditory channel in speech, thus, speech is the universal default channel for materialization of the language system, which influences the packaging of the message. Linguistic units are packaged to fit in intonation patterns. A sentence is organized to fit in a single prosodic contour. The boundary between a main clause and a complement clause is marked by a pause. Because vocal signals are ephemeral in nature, they require high speed of articulation and comprehension, which imposes elevated demands on cognitive functions and articulation during processing. This imposes restrictions on the length and the semantic and grammatical complexity of the utterances, ultimately shaping the characteristics of the language system. Gestural signs, like speech sounds, are ephemeral signs which makes them naturally suited for face-to-face communicative interactions, parallel to spoken dialogues.

The written text demands very differently organized linguistic message, composed of discrete and visibly distinguishable basic units, e.g. the letters of the alphabet or characters (in other types of writing systems) which are spatially organized. Letters are static and atemporal. This encourages the perception of discreteness of linguistic elements. Written texts are organized according to strict rules and conventions of grammatical correctness and conventions of punctuation. Importantly, written texts are organized around the sentence as a basic unit which encodes a complete thought by clearly defined grammatical categories and rules of grammaticality. The written text is self-contained, stands alone, independent of immediate context, communicative and extralinguistic, which demands detailed and explicit

exposition of the ideas it intends to convey and makes it well suited for externalization of code-like language system. As the written text is static and permanent, unlike speech, which is ephemeral and not recoverable, it facilitates the intellectual, argumentative and informative functions of language. It encourages the dissemination and accumulation of knowledge across geographic and temporal boundaries.

Thus, there is a consistent overlap between the channel of externalization and language functions, and by association, the language subsystems.

In this sense language evolution can be seen as coordinated mutual adaptation between modality and the language system.

4. Increment of semantic complexity, a catalyst for the evolution of language

The evolution of modern language from earlier, simpler forms is usually understood as imposition of grammatical rules on protolanguage, i.e. evolution of form. That said, if one is to define evolution in darwinian terms and claim survival advantages, one cannot but recognize that language must contain information about the environment as a reflection of human conceptualization of the world. Such viewpoint is consistent with Cognitive Grammar (Langacker, 2008; Evans, Green 2006). And, as mentioned earlier, complex thought does not necessarily require complex grammar. Scientific inquiries, works of literature and philosophy, laws, etc. are written in languages of varying levels of grammatical complexity, e.g. Latin, English, Chinese, etc. Complex ideas can be encoded in both complex and simple grammatical forms and simple ideas can be encoded in complex grammatical means, i.e. the same ideas can be encoded in different language systems, which makes translation generally possible. Cultures differ in the amount and nature of concepts encoded in linguistic forms and the types of constructions they use.

From the perspective of language evolution the informative function of language is instrumental in distributing crucial information assuring superior survival benefits to its users. Given that, the difference between earlier, more primitive stages, marked by limitations of information encoded, and modern language, the evolution of language is better understood in terms of semantic complexity. In this context the emergence of complex language is better explained with increase in conceptual complexity and the pressure for expanding the shared semiotic universe (Schoenemann, 1999, 2005). This, semiotic/semantic perspective refocusses the goals in the study of language evolution from tracing the evolution of structure, as currently articulated by major perspectives, to tracing the evolution of meaning representation. From the very essentials of human semiosis encoded in primitive form-meaning pairings, as demonstrated by rudimentary systems, further evolutionary transformations of the language system leading to the formation of modern language are better explicable with augmentation of the number of meanings (out of the totality of concepts humans are capable of conceiving) encoded in linguistic forms, lexical and grammatical (Dor, Jablonka 2000).

And if one subscribes to the argument that human ontogeny is a condensed illustration of phylogeny, observation of language development in youngsters could inform our understanding of language evolution in human species. In this context some scholars have argued that “under

the pressure to expand the shared meaning potential, hominids must have crossed the same “semiotic milestones” children cross in order to become the fore-bearers of a complex multidimensional semiotic system we now know as human language. “ (Williams, Lukin, 2004, p. 5). Language diversity is then understood as a result of the different ways communities organize their language systems by forming patterns of semanticization, i.e. transformation of conceptual structure into semantic structure by imposing semantic roles on concepts and their further encoding in linguistic forms.

The demand for extended meaning representation is stimulated in part by social changes, e.g. increase of group size and elaboration of group's internal organization, which brings as a consequence increase in complexity of social interactions and with it information inequality and the demand for more and more detailed information.(Matthiessen, in Williams, Lukin, 2004, p. 74-75). Extended semanticization is known to prompt the formation of new lexical categories.

ex. noun >adjective : orange (N) >orange (Adj) blouse

noun > adposition: front (N) > in front of (Adp) the window

verb > adposition: concern (V) >concerning (Adp) the letter

adjective >adverb: sincere (Adj) apology > sincerely(Adv)apologize

More examples can be found in Heine, Kuteva 2007. Another way of expanding the number of lexical items is through metaphorical extension of pre-existing literal meanings of nouns and verbs (Lakoff, Johnson, 1980). Aitchison (1999) has shown that the names of human body parts are used to name newly discovered concepts for objects in the outside world and emotional and intellectual states, and space vocabulary is extended to name time concepts.

In sum, language evolution is better understood in terms of meaning representation as increase of the proportion of concepts encapsulated in linguistic forms out of the totality of concepts comprising human semiosis, a process termed “ expansion of the expressive envelope” of language (Dor , Jablonka 2000). Naturally, this extended semanticization is directly reflected in increase of the number of lexical words and indirectly in the formation of additional new types of meanings encoded in fine-grained grammatical categories. In addition, in modern languages semiotic enrichment includes stylistic diversification demanded by a large society with social stratification as an additional layer of “ variation in language in relation to (social) context” (Williams, Lukin, 2004, p.81).

4.1. Bio-cognitive adaptations for extended semantics

A large number of concepts, essential for human survival, are universally encoded in linguistic terms. As mentioned earlier concepts of basic aspects of the environment, e.g., geography, flora, fauna, basic human biology , psychology and sociology , thus, concepts pertinent to essential survival, are universally encoded in all languages (Wierzbicka 1992; Goddard, 2007). And these are universally encoded in lexical words with concrete meanings. It is logical to anticipate that they form the very foundation of language and mark its very

beginnings. In addition universal patterns of categorization are universally reflected also in the grammars of all languages as “ semantic categories which are reflected by grammatical complexities in natural languages belong to a very constrained subset of all the categories we use to think, feel and conceptualize the world” (in Dor, Jablonka, 2000, p.39) e.g. categorical distinction of animate vs. inanimate, male vs. female, singular vs. plural, near vs. distant, etc.

Importantly, the very essentials of human patterns of categorization and their representation in lexical and grammatical categories, are expected to be reflected in the language-relevant properties of the brain, and likely have become evolutionary targets given their obvious survival benefits. In this sense elevated encephalization, suggesting elevated cognitive potential for extended conceptual complexity, has prompted some scholars to argue that evolution has favoured capacities for fast and effective learning of a large lexicon of constructions. Ph. Lieberman (2002 and elsewhere) has demonstrated in comparative studies of human and ape brains that areas of the human brain engaged in memorization and recall of linguistic forms are significantly more pronounced, suggesting that although apes can learn human words and successfully use them to communicate, facilitation of abilities to manipulate a large lexicon of constructions has been an evolutionary target during human evolution.

“ the almost three fold increase in the volume of these structures (prefrontal cortex and cerebellum) and the basal ganglia, compared to chimpanzees, could have yielded the computational base and memory size necessary to rapidly learn and store the meanings of new words... the posterior human brain, which current studies suggest is critical for accessing words from the lexicon ...is disproportionately large in humans compared to apes.” (Liebermann, Ph. 2002, p. 52).

In sum, aspects of the human organism show adaptation favouring the retention and processing of a large arsenal of human concepts in linguistic forms.

5. The evolution of language , a solution to the imbalance between human cognition and physiology

The imbalance between human cognitive and physiological capacities is relevant to the study of evolution of language as it translates into imbalance between the rich and highly detailed human semiosis , encoded in linguistic forms and the limitations of human physiology to articulate these effectively and in timely manner, producing a bottleneck. To remind, the term “ bottleneck” is borrowed from the Chaos theory and is instrumental for explaining the formation of structure from disorder in inorganic matter. In general bottlenecks are resolved by forcing transformation via emergence of patterned internal organization in the substance affected by the bottleneck. In the context of the present article it means that the language system must undergo transformation for the purposes of encoding more meanings by using linguistic forms more efficiently. I envision such transformations to encompass three processes: **1.** transformation of motivated signs, i.e. icons and indices, into symbols by replacing the natural connection with a convention which at the same time opens ways for compositional representation , thereby expanding the number of potential meanings, **2.** transformation in the signal thereby replacing multimodal, energy inefficient communication with much more energy

efficient speech, translated in physiological adjustments by evolving extended speech capacities, **3.** evolution of complex grammar as formation of abstract grammatical categories and principles of their organization into multilevel hierarchical structures as a compact representation of meanings.

5.1.Symbolization as efficient packaging of extended conceptual complexity

Words are generally assumed to have originated as symbols since the beginning of language. That said, some scholars argue that the earliest words were probably iconic signs, i.e. at the very beginning there must have been some kind of natural connection (iconic, indexical) between the meaning and form of the first words, facilitating their understanding by all communicators (Berling, 2005). Similarly, Bickerton hypothesizes that to assure adequate reception of the message by placing limits on the possible interpretations in situation of displaced reference the first words would have had to be iconic signals, e.g. imitations of natural sounds (Bickerton, 2009). Such arguments are supported by empirical findings of iconic gestures identified in bonobos (Genty, Zuberbuhler, 2015), demonstrating semiological continuity. Moreover, at early stage of language learning youngsters learn the names of animals by first associating them with their signature sounds, e.g. prelinguistic infants refer to a cat as “meow” before learning the word *cat*, demonstrating an initial stage of iconicity in vocalizations. In addition, at initial stages of formation in ASL and the AI-SAYIID sign language iconic gestures dominate and only later gradually become simplified and lose that natural connection by transitioning into symbols, demonstrating a shift from icons to symbols in the gestural space.

And given the identifiable natural connection of form and meaning in recently emerged sign languages and in language development in ontogeny, and under the assumptions that our past has left traces in our present, one must extrapolate similar trajectory from motivated signs to symbols in the evolution of language. The history of writing systems from pictographic images to symbolic representation of various levels of abstraction, e.g. by letters of the alphabet , Chinese characters, etc. suggests that this is a general tendency in the evolution of communication.

Icons are identified as detailed and elaborate depiction of the properties of the concept portrayed . This, naturally, involves time and energy-consuming efforts by communicators, which can be tolerated given that the amount of information remains constrained. At subsequent stages as the demand for more information increases, the functions of iconic communication are extended, which stretches human physiology to its limits and reveals the inherent limitations of iconic communication. As iconicity becomes an obstacle to the elevated demands for speed and precision due to sloppier articulation in speech and gesturing in manual signs, this triggers an internal restructuring in the communication system by transformation of iconic signs into symbols by eliminating the details, not essential for the accurate material representation and recognition of the concept denoted. As a result the natural connection between signifiant and signifie is completely lost and the icons are replaced by symbols, a process similar to the formation of writing systems from pictographic drawings.

The evolution of form in linguistic signs has inevitably resulted in evolution of the meanings these represents. Symbols help in creating discrete concepts and sharpen semantic categories. It has been argued that the use of symbols has had a major role in defining the boundaries of the concept, as “ symbols define the world.. Previously fuzzy properties become sharper after symbolization. “ (Donald 1991, p.219). Thus, the formation of symbols alters both meaning and form of linguistic signs, on one hand by streamlining of form by eliminating unnecessary detailing , and on the other enhancing the boundaries of meanings in major contribution to the efficient encoding of the expanding conceptual universe. To note, iconicity retains some influence in modern, predominantly symbol-based, language as Haiman (1983), Langacker, R. (2008), Evans, Green (2006) argue for a role of iconicity in vocabulary and grammar.

In sum, a transformative change in signs from icons into symbols allows for precision in meaning representation and efficiency of energy expense by minimizing the physiological effort in articulation. It is a unique solution to the disparity of cognition and physiology by cultural evolution of signs.

The human ability to form and process symbols is explicable in Deacon (1997 and elsewhere) in terms of a human capacity for symbolic reference, a complex combination of various types of referential relationships among symbols as members of a symbolic system, between a symbol and its referent, and among objects in reality as perceived by the human mind. It is argued to be a product of evolution of development, in adaptation to cultural evolution of symbolic from motivated signs and the evolution of language. The frontal lobe, which contributed most significantly to the brain enlargement during human evolution by increasing in volume about three-fold, compared to the ape brain, is identified as the site of location. In addition, in Deacon, the human brain has evolved extensive memory and precision of vocal control, unique features which taken together make language possible. Similarly, Bouchard (2012) argues that the human brain has evolved innate capacities to form and process Saussurean signs.

In short, linguistic symbols have evolved as efficient vehicle for representing and sharing key aspects of human semiosis in linguistic forms which has transformed the human mind and body.

5. 2. Evolution of speech, a solution to disparity between cognition and physiology

5.2.1. Multimodal roots of spoken language

Multimodal communication is defined as using multiple aspects of human physiology, from manual gesticulations, to facial expressions, body posture, vocalizations for communicative purposes. Multimodal signals are demonstrated by many species as distant from us as insects, frogs, birds, squirrels, etc. as they combine and coordinate signals from multiple modalities which complement one another to compose a convincing message (Slokombe et all. 2011). Thus, one could extrapolate that across the living world multimodality in communication is the rule rather than the exception.

Not unexpectedly, multimodality is at the heart of primate signalling. The communication of primates, both in natural habitats and in captivity, is a composite signal, i.e., a combination of various components which act in concert to deliver the complete message (Slocombe, Waller

and Liebal 2011).Chimps in captivity recognize and produce visual, tactile, vocal signs in their communication with caregivers and combine alarm calls with gaze alternations, suggesting voluntary control over manual gesticulations and individual vocal organs, e.g. control of breathing, larynx, supra-laryngeal articulators and capacities for vocal learning (Perlman 2017). Macaques's communication is cross-modal as well suggesting that “communicative complexity may be less about how each single modality is used, and more about signal integration. “ (Slocombe Waller, Liebal, 2011, p. 540) .Given these findings the unimodal approach to primate communication is inadequate for understanding a complex and integrated system as studying each component in isolation ignores the influence of the rest. See also Boe, L.J.et all 2017)

Inadequate understanding of primate communication is clearly an obstacle for evolutionary linguistics, especially if one subscribes to the continuity argument, assuming continuity of language with pre-human communication, as I do. In fact a plausible argument envisions multimodal beginnings of human language (Donald,1991; Levinson ,Holler 2014; Waller, et all, 2013; Slocombe, et all, 2011), suggesting that simpler semantic content was transmitted initially with a variety of signals by the whole body via a mixture of referential and iconic vocalizations, gesticulations, facial expressions, etc.

The argument for multimodal communication by pantomime (Zlatev et all., 2020; Ziwczyński et all., 2021) as a precursor to face-to-face dialogues, spoken and signed, originally proposed by Donald (1991) is held in high regard by many scholars(Corballis, 2003, 2007, Dessalles, 2007). Neurological explanation for multimodal beginnings of language focusses on the mirror neurons (Arbib, 2002 and elsewhere). Multimodal communication continues to form a substantial part in face-to-face dialogues today(Kendon, 2014, chap.6 and elsewhere)

That said, multimodal signals have high energy demands to human physiology, complemented by the cognitive demands of language processing during direct dialogues, places constraints on the use of multimodality to the very basic communicative necessities, indispensable in prehistoric environments, e.g. informing about food sources and predators, organizing hunt, settling disputes, etc. In addition, the high energy demands of multimodal communication to the human organism adds to the energy demands from the purely biological functions of the body, e.g. digestion, motion, etc. which would overwhelm the organism and make the use of language by no means beneficial to the overall survival of the biological body and the species as a whole. This suggests that a multimodal communication system of much lesser semantic and structural complexity than language as we know it today would have been in existence at the time of language origins. It is reasonable to assume that rudimentary subsystems, identified earlier, would provide an illustration of the very early forms of language, given the similarity of function.

Subsequent demands for increase of information content, prompted by the emergence of increasingly more complex social organization in larger populations, and with that increased frequency of interactions of expanding semantic complexity, stimulate the introduction of some grammatical sophistication , illustrated by inferential systems. The extended functions of language would have inevitably increased not only the cognitive but also the physiological demands to the organism which would have made the continuation of multimodal

communication prohibitively energy inefficient and even detrimental to the survival of its users and as such, unlikely target for evolutionary improvements. Following this line of thought, the focus of evolution on the physiological elaboration of the vocal channel as the most energy-efficient materialization of linguistic signs can then be seen as the most likely response to the physiological and kinetic limitations of the human body via adaptation for speech and the resulting universality of spoken language.

5.2.2 .The evolution of speech, a unique human adaptation

Speech is the preferred modality for linguistic communication in the overwhelming majority of languages. Although the linguistic signs can be materialized also through gestures (in sign languages) and writing, the term “natural language” implies a natural / biological association between language and speech. It is clear that spoken language precedes literacy by large margins, both evolutionary and developmentally, and sign language is learned only when the biological equipment enabling speech production and perception is damaged, as a substitute for speech, and not by choice.

Speech is a very unusual type of vocal signal in various ways: acoustically, physiologically, cognitively , developmentally. Acoustically, it is different from non-speech sounds, as demonstrated by acoustic equipment. Physiologically, it consists of rapid change of formants, produced by the rapid and coordinated movements of the organs in the vocal tract (Aaltonen, Uusipaikka, 2005; Pinker ,Jackendoff ,2009).

The fact that speech has evolved as the universal modality in human linguistic communication suggests that it is the most effective implementation of human physiology as a vehicle for materialization of human concepts in linguistic forms. For Lieberman “Without speech... complex language would not be possible”(Lieberman , Ph. 2008, p. 219,). See also Fitch (2010). The transformation from multimodal communication to speech, introduced initially at a behavioural level, was translated into anatomical and physiological adaptations by evolving a package of highly specific adaptations coordinated in a capacity for speech, e.g. highly agile vocal tract, able to produce sufficiently distinguishable discrete sounds, the human tongue with a unique shape and position in the oral cavity, the larynx , pharynx, the neck, face, coordinated in a vocal tract with unique physiology, species' specific breathing control, tied to speech production(Lieberman (2006, 2007).

Various authors have argued for continuity in the evolution of speech as various species have demonstrated vocal learning, instrumental in human speech development. Vocal learning happens in two stages, i.e. vocal imitation, or the ability to reproduce a perceived signal, and an ability to use this signal creatively in new contexts. Vocal imitation is uncommon in mammals, except for some marine species, and common in birds, bats and elephants (Hilliard, White, 2009; Petri, Schraff, 2011, p. 2125). The most relevant to language learning is the case of vervet monkeys, whose calls are mostly innate, although there is some learning involved as young vervet monkeys must learn to use their vocalizations in the appropriate context (Hilliard, White 2009). That said, humans have a rare capacity for complex vocal imitation to discriminate and reproduce series of vocalizations grouped in syllables, suggesting difference

in kind (Fitch 2010). A broad evolutionary continuity, suggested by the frame-content theory of speech evolution, argues that lip smacking gestures, an instinct, in primates were co-opted for the formation of the syllable and its most basic version as CV (Davis, MacNeilage (2004) MacNeilage (1998). Ralph-Axel Muller (1996) suggests that the emergence of the syllable as a linguistic unit can be traced evolutionarily to the “rhythmic jaw oscillation” and the rhythmic nature of the movements of the body-extremities. An evolutionary process of exaptation, i.e. by which a trait assumes a secondary adaptation for a new function, different from the original one, is envisioned as explanation for the for the new role of instinctive motion in speech production. He also suggests that this “general motor alphabet” (not specific for any group of muscles and comparable across species), when applied to the human vocal tract, yields the set of phonemes found in human languages.” In addition, he also claims that vowels originate from neonate “comfort noises” and trace the syllable back to “rhythmic jaw oscillation” analogous to rhythmic movements in the extremities.” (Muller, *ibid.*). Lieberman argues for continuity in the evolution of speech and attributes it to exaptation, in response to the increased demands of vocal communication. The finding that the FOXP2 gene, a major contributor to the phenotypic development of agile oral movements facilitating vocal communication, is present in Neanderthal genome, supports the argument that Neanderthals were capable of some form of speech (for more see Lieberman, 2007 and elsewhere). As per Lieberman the speech apparatus has been a distinct evolutionary target and one of the peculiarities of human speciation. Notably, although the capacity for speech has fundamental innate properties in the formation and configuration of human physiology, its proper functioning in communication requires a learning period during development, e.g. evolution has supplied human development with babbling, a unique period of training for speech, facilitating its development early in life. In a similar vein Oller, Griebel (2021) attribute a crucial role to the vocal dexterity and functional flexibility of the human vocal organs, unparalleled among the Homo branch, allowing the use of vocal signs in different contexts with different illocutionary interpretations, and emergent early in infant development.

Significantly, despite ample evidence for continuity, most clearly detected in Neanderthals, humans have evolved unique speech capacities for quantal vowels /i,a,u/, which are beyond the articulatory capacities of Neanderthals. (Lieberman, Ph. 2006 and elsewhere). Given that Neanderthals are also known to have limitations in the complexity of the toolkit, one can anticipate that the lack of advanced speech capacities is due to lack of evolutionary advantage for encoding semantic complexity. Thus, limitations in speech capacities can be used as indirect evidence for conceptual and semantic limitations.

Turning this argument on its head, the precision of articulation required for quantal vowels together with the demonstrable superiority in diversity and complexity of toolkit, can reasonably be interpreted as an indirect evidence for elevated conceptual and thus, semantic complexity of early humans. Thus, the phylogenesis of elevated speech capacities can reasonably be interpreted as adaptation to increased semantic complexity. It is a demonstration that human physiology adapts to human cognition by evolving articulate speech, an energy-efficient vehicle to meet the increased cognitive capacities and communicative demands. In sum, transformation of multimodal communication to speech has also lead to co-adaptation of the human organism by evolving exquisite speech capacities as a solution to the imbalance

between human cognition and physiology.

5. 2. 3. Speech influences the language system

It is consistently stated by the generative approach that the language system is immune from influence of its material instantiation, given the spatial and functional isolation of the grammar algorithm. Nevertheless, a fundamental principle of biology states that a biological property must be tailored to its function. Moreover, biological properties are only detectable by their functions. The generative approach has also consistently argued that the primary function of language is organizing the human conceptual system. That said, the only clear demonstration of the internal organization of human thought is by making it explicit in communication. The language system is only made accessible by its externalization via signs, which makes the nature of the signs a factor in the organization of the system. In fact, it is not difficult to notice that the semantics of a sentence materialized in spoken dialogues and written texts differ significantly. The organization of spoken dialogues is influenced by the ephemeral nature of vocal signals and the specifics of speech as a stream of discrete sounds emitted in sequence in addition to intonation and stress as markers of constituent structure. Thus, the nature of vocalizations imposes temporal restrictions on the speed of the information exchanged. In addition, linguistic forms are altered to fit the energy demands of the speech organs. Shorter forms which are often easy to pronounce, are used with high frequency in all languages as a universal tendency (Zipf, 1949). Speech constrains the length of the utterance, due to the natural limitation the speed of pronunciation imposed by the articulatory organs and limitations of the human episodic memory to retain information, restricted to 5-7 units at a time. Thus, speech imposes its influence on the language system, both in meaning and in structure.

In written texts, on the other hand, the organization of sentences is influenced by the nature of writing which affords stable, reliable information, recorded in minute detail with extensive demand for cognitive power, although with minimum physiological expense. Importantly, written language requires extensive training not only in the respective writing system, but also in elaborate grammatical code, imposing substantial cognitive demands, which explains the fact that throughout history it has been restricted to selected few, naturally constraining the use of complex language. To remind, mass literacy is a very recent phenomenon, prompted by the invention of the printing press in 14th century. In contrast, all humans learn and use with ease spoken language with minimum cognitive and physiological effort, which covers successfully the communicative demands of all human communities, and which, despite its deficiencies, provides adequate balance of effort and reward, justifying its universality. Thus, the physiology of speech influences language use which, in turn, influences the language system.

5. 2.4. The physiological demands of speech in spontaneous dialogues

Casual conversations among individuals with close social ties occupy about 20% of all waking time of humans (J. L.Dessalles, 2007). Spoken dialogues are universally the preferred form of language use. And although I do not have the precision of numbers as to the amount of

calories speech demands form the human body, it is clear that it is much less than multimodal communication demands for the same message. Speech is ideally suited for casual conversations among close friends and relatives where much information is shared and remains implicit. Direct linguistic interactions are conducted in dialogues where the cognitive and physiological effort is distributed among multiple participants as they take turns. Moreover, conversations are interspersed with stretches of silence, which often is a sign in itself and contributes to the overall meaning of the conversation, making the message even more energy-efficient for the speech organs. In addition, the fact that spoken conversations are conducted face-to-face also helps to infer and clarify the intended meaning by the facial expressions and/or body signals of the participants. In this way the physiological load is distributed among multiple aspects of human physiology, further reducing the energy demand for speech organs. Participation in casual conversations most often accompanies leisurely activities which presupposes relaxed atmosphere where expectation for semantic and grammatical accuracy and physiological demands for precision of articulation are relatively low as grammatical inaccuracies, conceptual vagueness, incomplete phrases, incomplete and often inaccurate pronunciation are tolerated and usually inconsequential for the accurate comprehension of the message.

Narration in spoken form, as long spoken monologues, e.g. speeches, lectures, etc., on the other hand, imposes a much higher demands, both cognitive and articulatory, which explains the fact that these are rare communicative events as they cause exhaustion to the narrator. Physical efforts in speech production are significantly reduced by the invention of writing, where the physiological cost is minimized by replacing it by the less energy-demanding visual channel, making it the most energy-efficient vehicle for dissemination of information.

On the other hand, one can reasonably anticipate that the production of signs in sign language has higher energy demands on physiology. And again, although I cannot tell precisely how many calories the body utilizes in the articulation of sign languages, the physiological demands of multimodal signs are comparable to those of everyday physical activities, e.g. gardening, cooking, etc. And by a general principle of biology, the biological body imposes limitations on physical activities. By the same principle the high kinetic demands of manual signs limit the use of sign language to short conversations. Prolonged conversations would have excessive physiological demands on the hands and the entire body, which would compromise the communicative process and result in significant simplification of the signs which would favour the sender, but compromise its expressivity and create ambiguity which inevitably would create more obstacles in comprehension for the receiver. The physiological demands on multimodal communication make it not suitable for long sentences with complex embedded structures and especially for narrative. This also explains the fact that speech and not sign language is the default channel, as the latter is only used when the vocal-auditory channel is unusable, as a solution of last resort, not as alternative to speech.

5. 3. Evolution of complex grammar as technology for packaging of human semiosis : emergence of grammar from meaning

Human concepts are organized into structured combinations and form propositions. Language

is a form of encapsulating human semiosis in linguistic forms. The generative paradigm, still influential in current linguistic thought, views the grammar of natural language as a systematic alignment of Language of Thought, an abstract system of human reasoning, with the grammar algorithm, or UG, another highly abstract system of linguistic categories and principles. Thus, both human thought and its articulation in language are envisioned as abstractions, i.e. insulated from human experiences.

Alternatively a semiotic perspective within the usage-based paradigm defines language as a system for encapsulating human semiosis in linguistic forms, a vision articulated by Williams and Lukin (2004) who argue that the formation and use of concepts, “the making of meaning” is the cornerstone in the emergence of language, in both ontogeny and phylogeny. Here the emergence of language both in the species and in the individual mind is viewed as a process of meaning formation and expansion.

This vision of human mind and language is supported by empirical studies, e.g. as per Pulvermuller (2005, 2018) linguistic meanings are grounded in reality as experienced from anthropocentric perspective. Moreover, the representation of the language system in the human mind in terms of abstractions cannot be justified in evolutionary terms except for its functionality, as a direct and significant contribution to the evolutionary success of its bearers, most clearly in its role of facilitating the dispersal of information about reality, crucial to survival. Thus, linguistic meanings cannot be devoid from human experience.

To note, reality is organized in hierarchies, from subatomic particles, to atoms and molecules, from single-cell organisms, to multicellular bodies, to societies, reflected in human semiosis and its representation by language. In this sense the expansion of human semiosis as a signature development in human evolution demands adjustments in the language system. Schoenemann (1999, 2005) argues that an extended lexicon, identified as an intermediate stage in the emergence of complex language, leads naturally to the emergence of hierarchical structure in syntax, as an accurate description of the structured organization of nature as encoded in linguistic forms, further followed by emergent syntax, evolved as a result of increased semantic content.

“hierarchical organization is exactly what one would expect if we assume that language evolved from simpler phrases to more complex ones....Given that the world is hierarchically organized (or at the very least that we have evolved to conceptualize the world hierarchically) should we not expect the syntax of our languages to reflect this?”(T. Schoenemann, 1999. p.330)

In this context the formation of hierarchies in nature and in language is viewed as a result of a process of self-organization, a natural phenomenon, a demonstration of the general tendency in nature towards structure formation.

“On theoretical grounds we could expect complex systems to be hierarchies in a world in which complexity had to evolve from simplicity”(ibid,1999, p.329,)

A similar idea is articulated in the “Now-or-never bottleneck” hypothesis, which explains the formation of phrases and sentences i.e. multilevel grammatical hierarchies, with a process of “chunk-and-pass” triggered by limitations in processing and articulation of language during ontogeny, which, repeated by numerous generations of learners, results in transformation of the

language system (Christiansen, Chater, 2016).

The need for such structural transformations are explained by some with social transformations and associated with them alterations of the ways language is used (Givon 1979). Similarly, Croft (2003) refers to a more elaborate social organization and its demand for precision in communication and language as explanation for complex grammar, i.e. societies with better internal organization speak better structured languages. Thus, a general tendency of capturing physical and social reality in ever increasing details are necessarily reflected in increasing conceptual and semantic complexity, ultimately encoded in elaborate grammatical machinery.

That said, although many languages choose to encode the exquisite detail of human semiosis in abstract grammatical forms and structures in search for communicative efficiency, this is by no means a universal tendency. And although the very essentials of human semiosis are universally captured by the very essentials of linguistic structure, as demonstrated by the rudimentary subsystems, languages differ widely in their choices of grammatical categories and principles of organization as tools for encoding the minute details of human knowledge, e.g. the same semantic content can be verbalized by language systems of varying degrees of grammatical complexity, as translation demonstrates. Thus, grammars are emergent communicative technologies, formed by communities in search for effective packaging of human semiosis for the purposes of communication.

At the same time the conceptual complexity of the human mind remains richer than what the language system can encode, i.e. the overlap between conceptual and semantic structure is only partial as only a subset of the conceptual arsenal is selected for representation in linguistic signs by lexicon and grammar (Jackendoff, 1990). Similarly, in Evans, Green (2006) “ while semantic structure must, to some extent at least, reflect conceptual structure, and while semantic structure can be thought of as a subset of conceptual structure, a system of lexical concepts selected for expression in language, the relationship between conceptual structure and semantic structure is...complex and indirect.” (Evans, Green, 2006, p.193). Thus, the mismatch between the human body and mind remains, although alleviated by the evolution of semantic and structural complexity of language .

5. 3.1.The evolution of grammar : from topic-comment to sentence structure

In current linguistic thought the evolution of language is pictured by some as transformation from protolanguage, here termed rudimentary system, into code (Bickerton,1984 and elsewhere) and by others as transformation from meaning-based system, or” pragmatic mode” into structure-based “syntactic mode” (Givon 1979 and elsewhere). The former is organized around information structure, where the information is divided in old and new, topic and comment and ordered according to pragmatic principles: the old precedes the new, abstract grammatical forms and structures are almost absent and extensive use of intonation and stress in marking the new/old contrast and potential ambiguity is resolved by the participation of non-linguistic body signals during participation in direct dialogues. Pragmatic mode is well suited for its use in endogenous communication, e.g. Givon's“ society of intimates”, i.e. a close-knit community with strong social ties, where information equality reduces the need for its explicit

articulation in linguistic means.

Significantly, not all thought is communicated and what is communicated is not only thought but also emotions, cultural forms of etiquette, etc. and it is well known that some elements of grammar have no semantic underpinnings, thus, have purely structural function. Such are the obligatory assignment of features of gender to nouns with meanings of inanimate entities, e.g. in French, Spanish, etc. the use of “it” as a place holder for the subject position in English, double marking of negation in many languages, etc. The formation of these abstract elements are an add on as an additional layer of structure, needed to assure the structural integrity of the system with the transformation from inferential system into a code. It is a reflection of the demands of extended language use in exogenous communication in new historical circumstances in Givon's “society of strangers”, i.e. in a large society with pervasive information inequality which imposes new demands on communication where participants communicate at a distance of space and time and the role of the extralinguistic context is largely eliminated. This places the burden of successful communication almost exclusively on language and triggers internal restructuring of the language system by replacing a meaning-based system with flexible internal organization, allowing freedom of semantic interpretation, by a structure-based system of strict rules of interpretation i.e. the inferential system is replaced by a code. This process is termed as syntacticization(Givon 1979 and elsewhere). The tight structure makes for fast and effective transmission of the message, communication becomes more efficient, are attributed to changes in social structure and resulting increase in complexity of social interactions, brought by the transformation of tribes and chiefdoms into states with large populations over vast territories and complex social organization (Croft 2003). States are organized around a central government which results in establishment of standard dialect on which government documents are written. In sum, the evolution of complex language is understood as replacement of less systematicity and structure with more.

5.4. The invention of writing as proliferation of complex grammar and the code

The invention of writing systems at the dawn of the first civilizations can be interpreted as a cultural solution to the significant increase of conceptual and semantic complexity associated with the radical social changes, mentioned above, resulting in information inequality driven by emergence of trades and social stratification. The demand for fast and efficient dissemination of knowledge beyond the limitations of space and time, imposed by face-to-face communication in spoken dialogues, was resolved by the invention of writing and a switch from the vocal to the much more energy-efficient visual channel. The switch in the channel of externalization brought changes in key aspects of languages. The proliferation of writing and literacy had a transformative role as it facilitated and encouraged increase in language complexity, both in meaning and form, e.g. large vocabulary of constructions and extensive use of recursion of phrases and sentences which extended the use of narrative. An example is the case of subordination in Greek which was absent in Homer's oral poetry, but is clearly present in the classic Greek literature and philosophy(Newmeyer, in Wray, 2002, p.369)

In short, the invention of writing has had dramatic consequences for the language system and its transformation from inferential communication to linguistic code, which prompted the

extension of complexity in the language system. In addition, by replacing speech with visual signals the physiological burden is reduced to a minimum, dramatically increasing the potential for massive dissemination of information.

5.4.1 .Cognitive adaptations for writing

The transformations in the language system both in form and in function have triggered some adaptations in human cognitive development , e.g. the natural flexibility of the human mind allows “ cultural recycling of cognitive maps ”(Dehaene, Cohen 2007) by which the visual system in early age can be trained for processing written language. It is found that the area of the cortex dedicated to processing visual information with location and shape invariant across cultures has demonstrated some flexibility which, after extensive exposure to experience with writing develops a dedicated area for processing orthographic symbols, detectable in the brain of a normal human literate adult. Moreover, despite cultural differences in the shape of letters, universal features are detected, e.g. the letters are composed of three strokes on average (Y, N) which intersect in specific ways, demonstrating co-adaptation of writing systems to the human brain as biology and culture co-evolution. The human abilities for processing orthographic shapes and the primate's visual capacities for object and scene recognition have shown similarities, suggesting evolutionary continuity. To note, the study in focus is limited to individuals familiar with the Roman alphabet, although the claim that the world's writing systems are designed to fit the innate biases of the primate visual system, which limits cultural variation, remains valid.

6 .Increasing functionality through efficiency

The evolution of language is usually understood as a tendency towards increasing the complexity of the language system, both semantic and structural, as detailed representation of extended semiosis requires detailed structural instruments for its encoding. As a counterargument Bichakjian (1999) argues that changes in language systems aim at increase in efficiency, i.e. increase in functionality while decreasing complexity of linguistic forms. This can be interpreted as reducing physiological cost by packing up more meaning in linguistic forms, i.e. efficient packaging of information content. An illustration of evolution of modern European languages from Proto -Indo- European could be taken as a general tendency and provide additional angle into understanding the evolution of language. For example the substitution of glottalized / kʔ, pʔ, bʔ/ and aspirate /kh, ph, bh/ consonants by obstruents /k, p, b / resulting in reduction of phonological complexity , which effectively increases the number of possible syllables, i.e. potentially increasing the number of concepts encoded., while minimizing articulatory efforts. Similar examples in the evolution of vowels demonstrate the same tendency of increase in functionality and material simplification. In grammar simplification is noticed in morphology by eliminating dual number from the number category, reducing it to singular and plural. Similar reduction in gender category, initially consisting of three grammatical genders, i.e. feminine, masculine and neuter, reduced to two,

confirms the tendency of simplification and streamlining in grammatical categories. Further simplification is noticed as earlier aspectual distinctions in verb morphology, encoded in the dichotomy of perfect tense, marking a complete action vs. imperfect, for incomplete or unfinished action, are substituted by past, present and future tenses on the basis of temporal distinction. In sum, languages have evolved to represent increasingly more complex thoughts more efficiently by packaging more information while reducing the inventory of linguistic forms and simplifying their material representation.

Summary and conclusions

The article presents a vision of language and its evolution somewhat deviating from the focus in current language evolution research. First, although language is a systematic way of curating and encoding human semiosis, structure is focused on as the emblematic feature, while meaning is an afterthought. The focus on structure informs inquiries of bio-cognitive representation of language in the human organism, which predominantly features the brain and its cognitive functions. Consequently these preliminary assumptions frame inquiries of language evolution. In contrast the present article views language primarily as an expression of human concepts. In addition, here the language user is understood as body and mind and the contribution of human physiology in the materialization of cognitive processing in linguistic messages is viewed as an important contributing factor in shaping the language system and its evolutionary path.

The article argues that a quirk in human speciation, produced by unprecedented increase of cognitive power resulting in unparalleled extension of conceptual and semiotic complexity, disproportionate to the physiological abilities, creates limitations for human dexterity to materialize it adequately in signs. Further, the article argues that the disparity between cognition and physiology has produced tension resulting in a bottleneck, formed by the limitations of pre-linguistic multimodal communication to adequately encode and articulate the extended human semiosis. It is resolved, the article argues, by the formation of language here viewed as incorporating three subprocesses which transformed pre-linguistic communication into modern language : **a.** evolution of symbols from motivated signs, **b.** evolution of speech from multimodal communication, **c.** emergence of grammar for efficient packaging of meanings. In this sense language evolution is a process of elaborating the rich human conceptual universe to fit into the limitations of human physiology for the purposes of communication.

The human organism has adapted both cognitively and physiologically to the demands of language processing by evolving extended memory storage and speech capacities. Human cultures have invented symbolic language systems and writing to accommodate the need for efficient and precise communication of semantic complexity while exploiting the natural adaptability of human body and mind. The language system has evolved in tandem with the biological body and human communities by evolving compact and efficient ways for packaging human semiosis as a compromise in response to the disparity between human cognition and physiology.

The article merely points at the disparity between human cognition and physiology, leaving

discussions of its evolutionary explanation to scholars with relevant expertise. And by addressing viewpoints currently unexplored it aims to contribute towards a more detailed and profound understanding of the evolution of language.

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